

 SERVERLESS ENGINEERING ARCHITECTURE

Serverless Rewards Ingestion Platform

A highly secure, infinitely scalable Spring Boot on AWS Lambda ecosystem that converts raw receipt uploads into instant rewards.

Frictionless **Client Capture**

Native Device Lens: Frontend triggers smartphone camera arrays natively utilizing bounds.

Stream Packaging: Binary data is wrapped in secure Axios FormData streams to avoid memory footprint corruption on-device.

ID Token Integration: Cryptographically paired with encrypted Cognito JWT keys inside transport headers.



Serverless Emulation Layer

1. Infrastructure Pipeline

The raw multipart request is routed straight to an AWS Lambda Function URL. AWS packages the stream into an modern HTTP Payload Version 2.0 framework (`HttpApiV2ProxyRequest`) with absolute binary safety, preventing raw text layout corruption.

2. Framework Emulation

Inside our entrypoint, a custom serverless handler mimics an active Apache Tomcat servlet engine. It smoothly translates raw JSON metadata (path, headers, payload body) straight into standard Java Spring `DispatcherServlet` routes.

Asynchronous **Validation & OCR**



EXIF Audit

Streams image bytes through metadata extractors. Validates real smartphone camera signatures, completely preventing digital spoof injections.



AnalyzeExpense

Invokes AWS Textract synchronous models to extract structured line-items, merchant tax ID details, and calculated receipt totals without templates.



Self-Auditing UI

Loops over line-items to algorithmically audit quantitative sums against receipt values:

$$\sum \text{Qty} \times \text{Price} \approx \text{Total}$$

Cryptographic Persistence

Zero-Trust Data Protection: The backend computes unique cryptographic hashes (SHA-256) of `Merchant_Total_Date` to completely eliminate and reject identical double receipt uploads.

Atomic Integrity Locks: Initiates transactional row-locks (`FOR UPDATE`) on our remote Supabase instance, preventing wallet balances from state-race conflicts during parallel writes.

 STEP 7: DATABASE ROW LOCKING & REWARDS



Image Sources



https://cdn.prod.website-files.com/6835b9105537c7e5f8f738d4/690f3481a814ed3903731a1d_94269560.png

Source: www.receiptvault.com



https://png.pngtree.com/background/20250207/original/pngtree-abstract-data-security-and-encryption-background-picture-image_15967386.jpg

Source: pngtree.com